Engineering Statistics

Yili Hong

Chapter 4: Discrete Distribution

Outline

- Random variables:
  - Definition, PMF, CDF
  - Expectation and Variance
- Common discrete Random variables:
  - Discrete Uniform
  - **Bernoulli, Binomial**
  - Geometric
  - **Poisson**
  - Poisson-Binomial distribution

Random Variable

Random variable:

- A variable whose value is a numerical outcome of a random phenomenon.
- It assigns a numerical value to each outcome in a sample space.
- Notation
  - Random variable: uppercase letters
    - e.g. $X, Y, \bar{X}, \ldots$
  - Observed value: lowercase letters
    - e.g. $x, y, \bar{x}, \ldots$

Example

Flipping a coin has two outcomes: $\{H, T\}$.

- We want to assign a value to each outcome.
  - We are free to choose ourselves.
- We simply choose what will be useful for addressing the questions that need to be answered later on.

Example

There can be many choices, for example,

1. Let $X = \begin{cases} 1, \text{when we observe head} \\ 0, \text{when we observe tail} \end{cases}$

or we can also define it this way,

2. Let $X = \begin{cases} 1, \text{when we observe head} \\ -1, \text{when we observe tail} \end{cases}$

or even this way,

3. Let $X = \begin{cases} \pi, \text{when we observe head} \\ -3e^{-2.3}, \text{when we observe tail} \end{cases}$

$$\vdots$$

Why Do We Need Random Variables?

Random outcomes are often **not numbers.**

$$\text{Outcome} \quad \longrightarrow \quad \text{Random Variable} \quad \longrightarrow \quad \text{Number}$$

Example: Toss two coins

$$HH, \; HT, \; TH, \; TT$$

Define

$$X = \text{number of heads.}$$

$$HH \to 2, \qquad HT \to 1, \qquad TH \to 1, \qquad TT \to 0.$$

Why?

Numbers allow us to use mathematics to study randomness.

Types of Random Variables

Discrete

- Random variables that have a countable list of possible outcomes
  - e.g. the number of scratches on a surface, the number of cracks on a highway

Continuous

- Random variables that can take on any value in an interval of real numbers for its range.
  - e.g. length, pressure, temperature, time, voltage, weight

Probability Mass Functions

Probability distribution:

- A description of the probabilities associated with the possible values of the random variable.
- For a discrete random variable X, its distribution can be described by a function that specifies the probability at each of the possible values for X.

Probability Mass Functions

Let X be a discrete random variable with possible values $x_1, x_2, \ldots$, a **probability mass function** is a function such that

1. $f(x_i) \geq 0$,
2. $\sum_{i=1}^{n} f(x_i) = 1$
3. $f(x_i) = P(X = x_i)$

- NOTE: **The probability rules must hold**

Example

Let X be the face value of a fair die with possible values $\{1, 2, 3, 4, 5, 6\}$.

x	$f(x)$
1	$1/6$
2	$1/6$
3	$1/6$
4	$1/6$
5	$1/6$
6	$1/6$

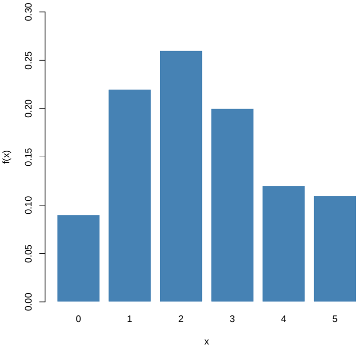
Example

Computer chips often contain surface imperfections. For a certain type of computer chip, 9% contains no imperfections, 22% contain 1 imperfection, 26% contain 2 imperfections, 20% contain 3, 12% contain 4, and the remaining 11% contain 5.

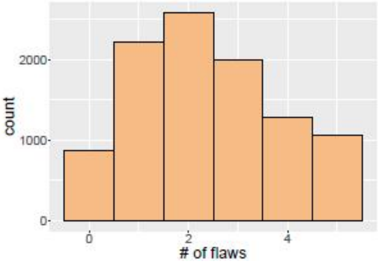
- Let X be # of imperfections in a randomly chosen chip.

x	$f(x)$
0	0.09
1	0.22
2	0.26
3	0.20
4	0.12
5	0.11

Example



PMF



Sample of 10,000

Remark

- For **discrete** random variables,
  - $P(X \leq x) = P(X < x) + P(X = x)$
  - $P(X \geq x) = P(X > x) + P(X = x)$
  - $P(X \leq x) = 1 - P(X > x)$
- e.g. X: # flaws on a randomly chosen computer chip
  - $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$
  - $P(X < 2) = P(X = 0) + P(X = 1)$

The Cumulative Distribution Function

The **cumulative distribution function** (CDF) of a random variable X, denoted as $F(x)$, is

$$F(x) = P(X \leq x), \quad \text{for any } x$$

- For **discrete** random variables X, $F(x)$ satisfies the following properties:

1. $F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$.
2. $0 \leq F(x) \leq 1$.
3. If $x \leq y$, $F(x) \leq F(y)$.

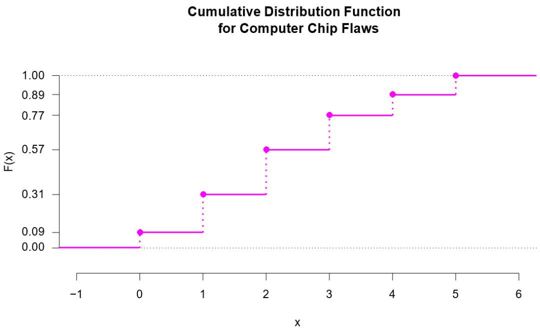
Example

Computer chips often contain surface imperfections. For a certain type of computer chip, 9% contains no imperfections, 22% contain 1 imperfection, 26% contain 2 imperfections, 20% contain 3, 12% contain 4, and the remaining 11% contain 5.

- Let X be # of imperfections in a randomly chosen chip.

x	$f(x)$	$F(x)$
0	0.09	0.09
1	0.22	0.31
2	0.26	0.57
3	0.20	0.77
4	0.12	0.89
5	0.11	1.00

Example


Cumulative Distribution Function
for Computer Chip Flaws
F(x)
1.00
0.89
0.77
0.57
0.31
0.09
0.00
−1
0
1
2
3
4
5
6
x

Remarks

- You will have noticed that the CDF is **right continuous** from the plot. This is always true.
- The cumulative distribution function can be written in the following way:

$$F(x) = \begin{cases} 0, & x < 0 \\ 0.09, & 0 \leq x < 1 \\ 0.31, & 1 \leq x < 2 \\ 0.57, & 2 \leq x < 3 \\ 0.77, & 3 \leq x < 4 \\ 0.89, & 4 \leq x < 5 \\ 1, & x \geq 5 \end{cases}$$

Example

1. What is the probability that we have 2 flaws or less?

$$P(X \leq 2) = F(2) = 0.57$$

2. What is the probability that we have more than 3 flaws?

$$P(X > 3) = 1 - P(X \leq 3) = 1 - F(3) = 1 - 0.77 = 0.23$$

Example

3. What is the probability that we have at least 2 flaws but fewer than 4 flaws.

$$\begin{aligned} P(2 \leq X < 4) &= P(2 \leq X \leq 3) \\ &= F(3) - F(1) \\ &= 0.77 - 0.31 \\ &= 0.46 \end{aligned}$$

- Note that we want to include 2.

Mean and variance of a random variable

Population mean and variance are often used to summarize a probability distribution for a random variable X.

- The population mean is a measure of the center or middle of the probability distribution
- The population variance is a measure of the dispersion, or variability in the distribution.

They do not uniquely identify a probability distribution, but they are simple and useful summaries of the probability distribution of X.

Mean of a Discrete RV

Let X be a discrete random variable.
Population mean:

$$\mu = E(X) = \sum_x x f(x), \quad f(x) = P(X = x)$$

where the sum is over all possible values of X.

- The population mean of X is also called the **expectation**, or **expected value**, of X and denoted by $E(X)$ or μ.
- We often call the mean **the first moment** of the random variable.

Expected Value of a Function of a Discrete Random Variable

Let X be a discrete random variable with PMF $f(x)$. Let $g(X)$ be a function of the random variable X, and hence is also a random variable.
The expected value of $g(X)$ is given by

$$E[g(X)] = \sum_x g(x) f(x)$$

Higher order expectation

- The rth moment of X is defined as $E(X^r)$
  - The 2nd moment: $E(X^2) = \sum_x x^2 f(x)$.
  - The 3rd moment: $E(X^3) = \sum_x x^3 f(x)$.

- The rth central moment of X is defined as $E((X - \mu)^r)$
  - The 2nd central moment: $E((X - \mu)^2) = \sum_x (x - \mu)^2 f(x)$.
  - In particular, the second central moment is called the **population variance** $Var(X)$.

Properties of Expectation

Let X be a random variable (discrete or continuous) and let c be any constant.

1. $E[c] = c$.
2. $E[cX] = cE[X]$.

- Note: All of the properties of expectation apply to both discrete and continuous random variables.

Variance of a Discrete RV

Let X be a discrete random variable.
Population variance:

$$\sigma^2 = Var(X) = E\left[(X-\mu)^2\right] = \sum_x (x-\mu)^2 f(x).$$

or using the property of expectation,

$$Var(X) = E(X-\mu)^2 = E(X^2) - \left[E(X)\right]^2 = \sum_x x^2 f(x) - \mu^2.$$

Population standard deviation:

- $\sigma = \sqrt{\sigma^2}$.

Properties of Variance

Let X and a random variable (discrete or continuous) and let c be any constant.

1. $Var[c] = 0$.
2. $Var[cX] = c^2 Var[X]$.

- Note: All of the properties of variance apply to both discrete and continuous random variables.

Common discrete random variables

- There are certain probability distributions that can be used to describe many common situations.
- We don't have time to cover all of them in this course, but we will focus on a few in this particular lecture.

Discrete Uniform distribution

Discrete Uniform distribution:

- A random variable X has a discrete uniform distribution if each of the n values, $\{x_1, x_2, \dots, x_n\}$, has equal probability. Then,

$$f(x_i) = \frac{1}{n}, \quad i = 1, 2, \dots, n$$

- Q. What is the probability that we randomly select number 1 among $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$?

Bernoulli Trials

A random experiment that can result in only one of two outcomes labeled as either as “success” or a “failure”, where the probability of success is p, and the probability of failure is $1 - p$, is called a **Bernoulli trial** with probability p.

$$\begin{aligned} P(\text{Success}) &= p \\ P(\text{Failure}) &= 1 - p \end{aligned}$$

Bernoulli distribution

For a Bernoulli trial, let $X = 1$ if the trial results in a success, and $X = 0$ if the trial results in a failure. Then,

$$\begin{aligned} P(X = 1) &= p \\ P(X = 0) &= 1 - p \end{aligned}$$

In this case, the random variable X is called a **Bernoulli** random variable with probability of success p.

Bernoulli distribution

Notation:

$$X \sim Ber(p)$$

- The symbol "$\sim$" means "distributed as" (or we say "follows").
- PMF:

$$f(x) = p^x (1-p)^{1-x}, \quad x = \{0, 1\}$$

Binomial distribution

Now we repeat **Bernoulli trials** n times. Let's assume that

- The total number of observations n is **fixed** in advance.
- Each observation has binary outcome:
  - **success** and **failure**.
- The outcomes of all n observations are statistically **independent**.
- All n observations have the same probability p of "success".

Binomial distribution

Let X be **the number of successes** in a series of n **independent Bernoulli trials**. X is known as a **Binomial** Random variable with parameters n and p. We denote

$$X \sim Bin(n, p)$$

- The number of successes $X \in \{0, 1, 2, \ldots, n\}$.
- The parameter n is the total number of trials.
- The parameter p is the probability of success on each trial.
- $Ber(p)$ is a special case of $Bin(n, p)$ with $n = 1$.
  - $X = \sum_{i=1}^{n} X_i$, $X_i \overset{indep}{\sim} Ber(p)$

Binomial Distribution in R

For $X \sim \text{Binomial}(n, p)$:

	R Function	Computes
PMF	`dbinom(x, n, p)`	$P(X = x)$
CDF	`pbinom(x, n, p)`	$P(X \leq x)$

Example: $X \sim \text{Binomial}(10, 0.3)$

```
# P(X = 4)
dbinom(4, size=10, prob=0.3)

# P(X <= 4)
pbinom(4, size=10, prob=0.3)
```

Example

A worn machine tool produces 1% defective parts. Let X be the number of defective parts in the next 3 parts produced. Then,

$$X \sim Bin(3, 0.01).$$

Example

Q. What is the probability that none of the parts are defective?

$$
\begin{aligned}
& P(\text{none of the parts are defective}) \\
= \quad & P(X = 0) \\
= \quad & P(\text{1st is not defective} \cap \text{2nd is not defective} \cap \text{3rd is not defective}) \\
= \quad & P(D^C)P(D^C)P(D^C) \\
= \quad & 0.99 \cdot 0.99 \cdot 0.99 \\
= \quad & 0.97
\end{aligned}
$$

Example

Q. What is the probability that exactly one of the three parts are defective?

$$
\begin{aligned}
& P(X=1) \\
= \;& P\big[(D \cap D^C \cap D^C) \cup (D^C \cap D \cap D^C) \cup (D^C \cap D^C \cap D)\big] \\
= \;& P(DD^CD^C) + P(D^CDD^C) + P(D^CD^CD) \\
= \;& P(D)P(D^C)P(D^C) + P(D^C)P(D)P(D^C) + P(D^C)P(D^C)P(D) \\
= \;& 3 \times 0.01 \times 0.99^2 \\
= \;& 0.0294
\end{aligned}
$$

Binomial distribution: PMF

Let X be the number of successes. Then,

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x},$$

- x: the number of successes, $x = 0, 1, 2, \ldots, n$
- n: the total number of trials.
- p: the probability of success on each trial.

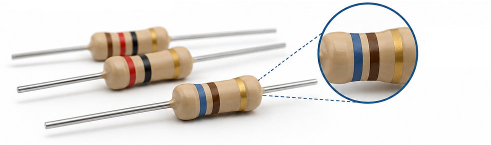
```
dbinom(x, n, p) # PMF
pbinom(x, n, p) # CDF
```

Example: Resistor Quality

Suppose 5% of resistors produced by a manufacturer are **out of specification.** We randomly select 10 resistors.
Let X be the number of out-of-specification resistors in the sample. Then

$$X \sim Bin(10, 0.05).$$

- Q. What is the probability that exactly two of the 10 resistors are out of specification?



Binomial distribution: Mean and Variance

Let $X \sim Bin(n, p)$. Then,

$$\begin{aligned} E(X) &= \mu = np \\ Var(X) &= \sigma^2 = np(1-p) \\ \sigma &= \sqrt{np(1-p)}. \end{aligned}$$

Example: Resistor Quality

Q. What is the expected number of out-of-specification resistors in a sample of 10 resistors?

$$\mu = np = 10 \times 0.05 = 0.5$$

Q. The population variance and population standard deviation are

$$\sigma^2 = np(1-p) = 10 \times 0.05 \times 0.95 = 0.475$$

$$\sigma = \sqrt{0.475} = 0.689$$

Geometric Distribution

Assume we conduct a sequence of independent Bernoulli trials each with success probability p.

Let X be **the number of trials** up to and including the first success.

Then X is said to have the **geometric** distribution with parameter p. We denote

$$X \sim Geom(p).$$

Geometric Distribution

- PMF:
$$f(x) = p(1-p)^{x-1}, \quad x = 1, 2, \ldots,$$
- CDF:
$$F(x) = 1 - (1-p)^{x}, \quad x = 1, 2, \ldots$$
- Mean and variance:
$$E(X) = \frac{1}{p}, \quad Var(X) = \frac{1-p}{p^2}.$$

Example

Suppose you are playing basketball, and you want to model the number of attempts it takes for a player to make their first successful free throw. Assume that each free throw attempt is independent of the others, and the probability of making a free throw is 0.7.

Q. What is the probability that it takes three attempts until the first success of free throw?

$$X \sim Geom(0.7) \Rightarrow P(X = 3) = 0.7 \times 0.3^2 = 0.063.$$

- The expected number of attempts is $E(X) = 1/p = 1.429$.

Negative Binomial Distribution

The **negative binomial** distribution is an extension of the geometric distribution. Assume independent Bernoulli trials each with success probability p. Then let X denote **the number of trials** up to and including the rth success. Then X has the negative binomial distribution with parameters r and p, written

$$X \sim NB(r, p).$$

Poisson distribution

A **Poisson** distribution describes the **count** X of occurrences of an event in **fixed, finite intervals** of time or space when the occurrences are all **independent.**
Let λ be the average number of occurrences for a specified interval of time or space. Let X be the number of occurrences in the same unit of time/space. Then X is said to have **Poisson** distribution.

$$X \sim Pois(\lambda).$$

Poisson distribution

The Poisson probability distribution of observing x occurrences is:

$$f(x) = P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!},$$

where $x = 0, 1, 2, \ldots$ and $\lambda > 0$.

```
dpois(x, lambda) # PMF
ppois(x, lambda) # CDF
```

- Mean and Variance:

$$E(X) = \lambda, \quad Var(X) = \lambda.$$

Example

The number of hits on a certain website follows a Poisson distribution with a mean rate of 4 per minute.
Q. What is the probability that 5 visitors view the page in a given minute?

Poisson distribution with t units

If λ is the mean number of events in **one unit** of time/space, then the mean number of events in t units of time/space is λt, hence

$$Y \sim Poisson(\lambda t)$$

Example

The number of occurrences receiving emails follows a Poisson distribution with a mean rate of 4 per minute.
Q. What is the probability that fewer than 2 emails are received in a period of 30 seconds?

Poisson approximation

Suppose we have a situation where n is large and the probability of success p is small:
A mass contains 10,000 atoms of a radioactive substance. The probability that a given atom will decay in a one-minute period is 0.0002. Let X = # of atoms that decay in one minute. Each atom can be thought of as a **Bernoulli** trials, so

$$X \sim Bin(10000, 0.0002).$$

with $E(X) = (10000)(0.0002) = 2$. Suppose we wish to find $P(X = 3)$:

$$P(X = 3) = \binom{10000}{3} 0.0002^3 0.9998^{9997} = 0.180465$$

Poisson approximation

It can be shown that if n is large, and p is small, then if $\lambda = np$,

$$
\begin{aligned}
& \binom{n}{x}\left(\frac{\lambda}{n}\right)^x\left(1-\frac{\lambda}{n}\right)^{n-x} \\
= \quad & \lambda^x \underbrace{\binom{n}{x}\left(\frac{1}{n}\right)^x}_{\to\frac{1}{x!}} \underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\to e^{-\lambda}} \underbrace{\left(1-\frac{\lambda}{n}\right)^{-x}}_{\to 1} \\
\to \quad & e^{-\lambda}\frac{\lambda^x}{x!}.
\end{aligned}
$$

Example

For the radioactive decay problem, we see that we can approximate Binomial distribution to Poisson distribution since $n = 10000$ and $p = 0.0002$. In this case, the mean observed decayed atoms is $\lambda = np = 2$.
Using the PMF of the Poisson distribution, the probability that we will observe 3 atoms decay (out of 10000) in a one-minute period of time is

$$P(X = 3) = e^{-2}\frac{2^3}{3!} = 0.180447$$

Other Common Discrete Distributions

There are many other useful discrete probability distributions:

- **Negative Binomial Distribution**
  Number of trials needed to observe a specified number of successes.
- **Hypergeometric Distribution**
  Number of successes when sampling **without replacement.**
- **Multinomial Distribution**
  Extension of the binomial distribution to more than two possible outcomes.

Case Study: Learning a New Distribution with AI

Suppose you encounter the term **Poisson binomial distribution** in an engineering or computer science problem, but we have not covered it in class.

Ask AI

I know the Bernoulli and binomial distributions.
What is the Poisson binomial distribution?
Explain it by connecting it to what I already know.

Key idea

AI can serve as an on-demand tutor for learning new statistical concepts.

From Binomial to Poisson Binomial

- Independent Bernoulli random variables: $X_i = 1$ for success and $X_i = 0$ for failure.
- **Binomial distribution:**
  Same probability for every trial:

$$P(X_i = 1) = p. \quad X = \sum_{i=1}^{n} X_i \sim \mathrm{Binomial}(n, p).$$

- **Poisson binomial distribution:**
  Probabilities can be different:

$$P(X_i = 1) = p_i, \qquad X = \sum_{i=1}^{n} X_i.$$

Binomial: same p vs. **Poisson binomial:** different p_i

Computer Science Example: Server Failures

A distributed computing system has five **independent servers.** Because the servers have different hardware, ages, and workloads, their probabilities of failure during the next 24 hours are

$$p = (0.01, 0.02, 0.05, 0.08, 0.10).$$

Let X = number of servers that fail.

Question

What probability distribution should we use to model X?

Why Not Use a Binomial Distribution?

- For a binomial distribution, we need

$$X = \sum_{i=1}^{n} X_i,$$

where the X_i's are independent Bernoulli random variables with the same probability of failure:

$$p_1 = p_2 = \cdots = p_n = p.$$

- Our servers have different failure probabilities:

$$0.01, \quad 0.02, \quad 0.05, \quad 0.08, \quad 0.10.$$

The probabilities are different, so X is **not binomial.**

Ask AI for Help

We know the binomial distribution, but this problem is different.

Ask AI

I have 5 independent servers with failure probabilities

$$0.01, \ 0.02, \ 0.05, \ 0.08, \ 0.10.$$

I want to model the total number of servers that fail.
I know the binomial distribution. Can I use it here?
If not, what distribution should I use? Explain it by connecting it to the binomial distribution.

AI Explains the Poisson Binomial Distribution

- The **Poisson binomial distribution** is the distribution of

  $$X = \sum_{i=1}^{n} X_i,$$

  where the X_i's are independent Bernoulli random variables with

  $$P(X_i = 1) = p_i.$$

  The p_i's do not need to be equal.
- Mean and variance:

  $$E(X) = \sum_{i=1}^{n} p_i, \qquad \mathrm{Var}(X) = \sum_{i=1}^{n} p_i(1 - p_i).$$

- When all $p_i = p$, it reduces to the binomial distribution.

Ask AI: How Do I Compute It in R?

Is there an R package for the Poisson binomial distribution?
AI may suggest the `poibin` package.

```
install.packages("poibin")    # run once
library(poibin)

p<-c(0.01,0.02,0.05,0.08,0.10)
```

The package provides functions similar to other distributions:

- `dpoibin()` — probability mass function
- `ppoibin()` — cumulative distribution function
- `qpoibin()` — quantile function
- `rpoibin()` — random number generation

R Computing: Server Failures

Let X be the number of servers that fail, with

$$p = (0.01, 0.02, 0.05, 0.08, 0.10).$$

```
library(poibin)

p<-c(0.01,0.02,0.05,0.08,0.10)

# P(X=2)
dpoibin(2,pp=p)

# P(X<=2)
ppoibin(2,pp=p)

# P(X>=1)
1-ppoibin(0,pp=p)

# Entire probability distribution
x<-0:5
prob<-dpoibin(x,pp=p)
prob
```

Visualize and Verify

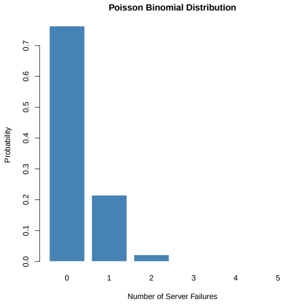
Plot the probability distribution:

```
x<-0:5
prob<-dpoibin(x,pp=p)

barplot(prob, names.arg=x,
        xlab="Number of Server Failures",
        ylab="Probability",
        main="Poisson Binomial Distribution")
```

Check your results

- `sum(prob)` should be 1.
- Mean should equal `sum(p)`.
- Variance should equal `sum(p*(1-p))`.



Chapter Summary

- **Use random variables to represent random outcomes numerically.** This allows us to study randomness mathematically.
- **Use probability distributions to describe uncertainty.** For discrete random variables, the PMF gives individual probabilities and the CDF gives cumulative probabilities.
- **Choose a distribution based on how the random variable arises.** Bernoulli models a binary outcome; binomial counts successes; geometric counts trials until the first success; and Poisson counts events over an interval.
- **Use computing and AI to extend statistical knowledge.** R computes and visualizes distributions; AI can help learn unfamiliar distributions, but results should be understood and verified.